Review Report for "Constructive and Destructive Training Regimes: A Framework for Interpreting SGD Stability"

This paper presents a signal-inference framework to study the stability of SGD and its variants, including the minibatch, momentum and weight decay. The main results are some control on the norm of SGD iterates in terms of gradient energy, intra-batch alignment and cross-step interference. The paper also include some empirical analyses to show the training loss, training trajectories and empirical stability ablations.

Comments:

(1) While the paper gives some control on the norm of SGD iterates, the paper does not give an explicit bound of this norm in terms of the iteration number, step size and minibatch size. The paper just relates this norm to several other terms such as the accumulated gradient energy and intra-batch alignment. It seems that these analyses do not fully use the property that SGD is actually trying to minimize an objective function. In the literature, there are some existing analyses on the control of SGD iterates by martingale concentration inequalities, e.g., [1, 2], which show that SGD iterates stay inside a bound with a finite radius with high probability. However, the paper does not give discussions with these related works.

(2) The analyses in controlling the SGD iterates are standard. The work just expands the squared distance to decompose it to several terms. I do not see challenges in the analyses.

(3) The analyses impose restrictive assumptions. For example, above Eq (12) the paper assumes $G_{B_t}^\top G_{B_u} \approx 0$ for $|t-u| > 1$. This assumption is restrictive since it assumes the correlation is small beyond one iteration, which is unlikely to hold in practice. For example, if the step size is small, the correlation can still be very large even after a number of iterations. The paper does not give empirical support to this assumption.

(4) The paper estimates the distance between two successive batches in Eq (30). However, it is possible that the right-hand side in Eq. (30) is negative because it involves the subtraction of two terms. In this case, the distance is not well defined.

References

[1] Y. Lei. Towards better generalization bounds of stochastic optimization for nonconvex learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.

[2] S. Li and Y. Liu. High probability guarantees for nonconvex stochastic gradient descent with heavy tails. In *International Conference on Machine Learning*, pages 12931–12963. PMLR, 2022.